

Scalability & Future Plan

Date	1 July 2026
Team ID	XXXXXX
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Local deployment only	Limited accessibility	High
2	No user authentication	Basic security limitations	Medium
3	Crop recommendation only	Limited functionality	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports local users	Deploy on cloud platform
2	Data Storage	CSV dataset	Use MySQL or PostgreSQL
3	Performance	Single ML model	Optimize model and caching
4	Security	Input validation only	Add user authentication

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Cloud deployment	1–2 Months	Public accessibility
Phase 3	Fertilizer recommendation	2–3 Months	Better farming support
Phase 4	Plant disease detection	3–4 Months	Complete crop assistance